NGUMIMI KAREN IYORTSUUN

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Pattern Recognition Laboratory
Chonnam National University
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South Korea

RESEARCH INTEREST

- Cross-modal learning for mental health assessment
- Multimodal depression detection using EEG, audio, and text data
- EEG functional connectivity analysis for mental disorders
- AI-driven clinical decision support systems

EDUCATION

PhD (Integrated Master and PhD): March 2022 - Present

Chonnam National University, Gwangju, South Korea, Artificial Intelligence

Dissertation: Coupling Without Correspondence: Shared-Weight Learning for Depression Detection Across Unpaired, Subject-Disjoint Cohorts

Scheduled defense: November 2026 | **Expected graduation:** February 2027

BSc: University of Mkar, Mkar, Benue State, Nigeria, Computer Science August 2019

Thesis: Web-Based Library System: a case study of the University of Mkar

Advisor: Sever Kwaghbee

RESEARCH EXPERIENCE

Graduate Student March 2022 - Present

Chonnam National University, South Korea

Cross-Modal Mental Health Detection using EEG, ECG, Audio, and Textual Modalities

Supervisor: Professor Soo-Hyung Kim

- Preprocessed EEG, ECG, audio, and text data from multiple depression and anxiety datasets (DAIC-WOZ, PRED+CT, MODMA, EATD Corpus, Chinese multimodal, Physionet Spider dataset)
- Designed and implemented deep learning architectures for cross-modal depression detection using attention mechanisms and Siamese networks
- Performed EEG functional connectivity analysis using coherence and phase-locking value
- Conducted statistical validation and ablation studies for model performance evaluation
- Published 3 first-author SCIE papers (Healthcare, IEEE Access, IEEE Sensors Journal), 3 co-authored SCIE papers, and multiple conference papers with over 500 total citations (h-index: 6)

Research and Teaching Assistant March 2022 – Present

Pattern Recognition Laboratory, Chonnam National University

Advisor: Professor Soo-Hyung Kim

- Designed experimental pipelines for multimodal fusion using transformer-based architecture
- Implemented neural network training and optimization procedures
- Support course activities such as serving as a proctor during examinations
- Managed course results, handling and verifying examination grades
- Mentor graduate students on deep learning implementation and experimental design

MENTORING AND LEADERSHIP

Lab Leader February 2023 - Present

Pattern Recognition Laboratory, Chonnam National University | **Mentees:** 8

- Guide and onboard new graduate students in deep learning implementation, experimental design, and research workflows, and advise on their ongoing projects and academic progress
- Manage the lab website and serve as liaison between the department and lab members

Student Mentor March 2023 – December 2023

Chonnam National University | **Mentees:** 5

- Provided academic guidance to junior students on coursework, research direction, and graduate study

HONOURS AND AWARDS

Best Paper Awards

- “AI-Driven EEG Signal Processing: Advancements in Depression Marker Identification” - *14th International Conference on Smart Media & Applications (SMA2025)*.
- "EEG Statistical Analysis of Coherence Patterns in Depression and Machine Learning Classification," - *Annual Conference of Korea Information Processing Society (KIPS2025).*
- “MS-HRVNet: Multi-Scale Residual Learning for ECG-Based Anxiety Detection” – *Annual Symposium of the Korea Information Processing Society (ASK2026)*

Competition Awards

- Gold Prize, 2025 Industry-Academic Cooperation and AI Hackathon Competition (*HANA TALK translation system*)
- Encouragement Award, 2023 AI Industry-Academic Cooperation Project (D*ocument masking system*)

Scholarships and Fellowships

- Brain Korea 21 (BK21) Scholarship 2022–Present
- Industry-Academic Cooperation Foundation Scholarship 2022–Present
- Truth Scholarship (partial tuition) 2024
- Talent Development Center Scholarship 2023
- SRS Scholarship (full semester tuition) 2022

PUBLICATIONS

Journal Publications

N. K. Iyortsuun, S. Yeom, M. Jhon, H.-J. Yang, S.-W. Kim, J.-E. Shin, and S.-H. Kim, “A Frequency-Band Siamese Network with Dual Functional Connectivity Metrics for Subject-Independent EEG Depression Detection,” *IEEE Sensors Journal*, early access, 2026 doi: 10.1109/JSEN.2026.3737137.

N. K. Iyortsuun, S.-H. Kim, S. Yeom, H.-J. Yang, and A. Kim, “Event-Level Anxiety Detection from Short-Window Heart Rate Variability,” *Journal of Information Processing Systems*, accepted, 2026.

N. K. Iyortsuun, S.-H. Kim, S. Yeom, H.-J. Yang, A. Kim, "Enhanced EEG Coherence Analysis for Major Depressive Disorder: Cross-Frequency Ratios and Patient Heterogeneity," *The Transactions of the Korea Information Processing Society*, vol. 14, no. 10, pp. 785-795, 2025.

T. N. Nguyen, S.-H. Kim, B. G. Kho, N. T. Do, **N. K. Iyortsuun**, G. S. Lee, and H.-J. Yang, "Temporal variational autoencoder model for in-hospital clinical emergency prediction," *Biomedical Signal Processing and Control,* vol. 100, p. 106975, Feb. 2025.

H. H. Nguyen, **N. K. Iyortsuun**, S. Kim, H.-J. Yang, and S.-H. Kim, "Mental workload estimation with electroencephalogram signals by combining multi-space deep models," *Biomedical Signal Processing and Control,* vol. 94, p. 106284, Aug. 2024.

N. K. Iyortsuun, S.-H. Kim, H.-J. Yang, S.-W. Kim, and M. Jhon, "Additive cross-modal attention network (ACMA) for depression detection based on audio and textual features," *IEEE Access,* vol. 12, pp. 20479–20489, 2024.

K. N. Phan, **N. K. Iyortsuun**, S. Pant, H.-J. Yang, and S.-H. Kim, "Pain recognition with physiological signals using multi-level context information," *IEEE Access*, vol. 11, pp. 20114–20127, Feb. 2023.

N. K. Iyortsuun, S.-H. Kim, M. Jhon, H.-J. Yang, and S. Pant, "A review of machine learning and deep learning approaches on mental health diagnosis," *Healthcare*, vol. 11, no. 3, p. 285, Jan. 2023.

Submitted/Under Review

N. K. Iyortsuun, H.-J. Yang, M. Jhon, and S.-H. Kim, “Depression Detection with Shared-Weight Multi-Modal Coupling across Unpaired, Subject-Disjoint Cohorts,” *IEEE Journal of Biomedical and Health Informatics*, **submitted**, 2026.

Conference Papers

N. K. Iyortsuun, S.-H. Kim, S. Yeom, H.-J. Yang, and A. Kim, “MS-HRVNet: Multi-Scale Residual Learning for ECG-Based Anxiety Detection,” in Proceedings of the Annual Symposium of the Korea Information Processing Society (ASK), Gangneung, South Korea, May 2026.

H.-S. Vo, K. Kolmogortseva, **N. K. Iyortsuun**, H.-D. Vo, and S.-H. Kim, “GlobalizeEd: An Editable Multimodal Translation Pipeline for Cross-Lingual Academic Lectures,” in Proc. ACM Multimedia Asia (MM Asia), Hanoi, Vietnam, 2026, accepted.

N. K. Iyortsuun, S. Yeom, H.-J. Yang, S.-W. Kim, J.-E. Shin, and S.-H. Kim, "AI-driven EEG signal processing: Advancements in depression marker identification," in Proc. 14th International Conference. Smart Media & Applications (SMA), Istanbul, Türkiye, Aug. 2025.

N. K. Iyortsuun, S.-H. Kim, S. Yeom, H.-J. Yang, and A. Kim, "EEG statistical analysis of coherence patterns in depression and machine classification," in Proceedings of the Annual Conference of Korea Information Processing Society (KIPS), pp. 463–465, May 2025.

N. K. Iyortsuun, S.-H. Kim, and A. Kim, "Protecting privacy in the digital age: A novel approach for Korean ID card anonymization," in Proceedings of the Korea Smart Media Society & Korea Electronic Commerce Society Spring Conference, pp. 39–40, May 2024.

N. K. Iyortsuun, S. Yeom, S.-H. Kim, and A. Kim, "Phase locked value-based Siamese network for EEG depression classification," in Proceedings of the Annual Conference of Korea Information Processing Society (KIPS), pp. 549–552, 2024.

N. K. Iyortsuun, S.-H. Kim, H.-J. Yang, and A. Kim, "Artificial intelligence techniques in mental healthcare: A systematic mapping study," in Proceedings of the 10th Int. Conf. Big Data Appl. Services (BIGDAS), Jeju, South Korea, Nov. 2022.

SKILLS/TECHNICAL PROFICIENCIES

- **Programming & Frameworks:** Python, PyTorch, TensorFlow
- **Signal Processing:** EEG preprocessing and functional connectivity analysis (coherence, phase-locking value), audio feature extraction, time-frequency analysis
- **Machine Learning:** Transformer architecture, attention mechanisms, Siamese networks, cross-modal learning, multimodal fusion
- **Tools & Libraries:** Scikit-learn, MNE-Python, Librosa, Pandas, NumPy
- **Statistical Analysis:** Cross-validation, significance testing, ablation studies
- **Version Control:** Git, GitHub

PROFESSIONAL SERVICE

Peer-Review Activity:

- Reviewer, BMC Psychiatry (2 manuscripts reviewed)

PROFESSIONAL AFFILIATION

- Associate Member, Korea Information Processing Society 2024 - Present